King Saud University
College of Computer and Information Sciences
Department of Information Technology

IT 462:

Big Data Systems

Course Project

2st semester, 2025/26

Diabetes Health Indicators

Phase 4

Group#:	**3**		
Section#:	**2766**		
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1. Overview

In this phase, Spark SQL and DataFrame operations were applied to the preprocessed Diabetes Health Indicators dataset. The dataset was converted into a structured DataFrame and registered as a temporary SQL view, enabling efficient querying using standard SQL syntax within Spark.

This phase focuses on extracting structured insights using aggregation, filtering, and ranking queries to better understand the relationships between diabetes, lifestyle factors, and demographic attributes. Five non-trivial SQL queries were designed and executed, each targeting a specific analytical question.

2. DataFrame Creation and SQL View Registration

Before executing SQL queries, the preprocessed dataset was loaded into Spark as a DataFrame and registered as a temporary SQL view. This allows the dataset to be queried using standard SQL syntax within Spark.

```
scala> val df = spark.read
     |   .option("header", "true")
     |   .option("inferSchema", "true")
     |   .csv("/Users/jood/Downloads/Diabetes Preprocessed Data.csv")
     |
[    | df.createOrReplaceTempView("diabetes_data")
val df: org.apache.spark.sql.DataFrame = [Diabetes: int, HighBP: double ... 21 more fields]
```

Explanation:

- The dataset was loaded from a CSV file into a Spark DataFrame.
- The header option ensures column names are correctly recognized.
- The inferSchema option automatically assigns appropriate data types.
- The DataFrame was then registered as a temporary view named diabetes_data, enabling SQL queries to be executed on it.

3. SQL Queries and Analysis

Note on Age encoding: The Age column is an ordinal code (1–13) from the BRFSS codebook, not a literal age. Code 1 = 18–24 years, Code 2 = 25–29, …, Code 9 = 60–64, Code 10 = 65–69, Code 11 = 70–74, Code 12 = 75–79, Code 13 = 80 or older.

3.1. Diabetes Rate by Age Group

Analytical question: How does diabetes prevalence differ across age brackets?

SQL Query

```
scala> spark.sql("""
     SELECT
         Age,
         COUNT(*) AS total_individuals,
         SUM(CASE WHEN Diabetes > 0 THEN 1 ELSE 0 END) AS diabetic_individuals,
         ROUND(
             SUM(CASE WHEN Diabetes > 0 THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
             2
         ) AS diabetes_rate
     FROM diabetes_data
     GROUP BY Age
     ORDER BY Age
     """).show()
```

Insight

The results show a strong positive relationship between age and diabetes prevalence, increasing from about 3% in younger groups to nearly 49% in older groups. This indicates that age is a major risk factor for diabetes.

Query Output:

age	diabetes_rate
1	3.2%
2	4.7%
3	6.98%
4	11.37%
5	16.08%
6	21.92%
7	28.36%
8	33.06%
9	40.6%
10	46.8%

11	48.9%
12	46.8%
13	40.8%

3.2. Average BMI by Diabetes Status

Analytical question: Do individuals with diabetes have higher BMI values compared to non-diabetic individuals?

<table>
<tr><th>SQL Query</th><th>Insight</th></tr>
<tr><td>
<pre>
scala> spark.sql("""
     | SELECT
     | Diabetes,
     | COUNT(*) AS total_individuals,
     | ROUND(AVG(BMI), 2) AS average_bmi
     | FROM diabetes_data
     | GROUP BY Diabetes
     | ORDER BY Diabetes
     | """).show()
</pre>
</td><td>Individuals with diabetes have a higher average BMI than non-diabetic individuals, increasing from around 28 to over 31 in more severe cases. This confirms a strong link between higher body weight and diabetes risk.</td></tr>
</table>

Query Output:

Diabetes	total_individuals	average_bmi
0 (No Diabetes)	191,831	27.96
1 (Prediabetes)	4,748	31.14
2 (Diabetes)	32,921	32.26

3.3. High-Risk Individuals Count

Analytical question: How many individuals simultaneously exhibit multiple major diabetes risk factors?

<table>
<tr><th>SQL Query</th><th>Insight</th></tr>
<tr><td>

```
scala> spark.sql("""
     | SELECT COUNT(*) AS high_risk_count
     | FROM diabetes_data
     | WHERE HighBP = 1
     |   AND Smoker = 1
     |   AND PhysActivity = 0
     |   AND UnhealthyLifestyleScore > 2
     | """).show()
+---------------+
|high_risk_count|
+---------------+
|           1281|
+---------------+
```

</td><td>A total of 1,281 individuals were identified as high-risk based on combined factors including high blood pressure, smoking, lack of physical activity, and poor lifestyle scores. This demonstrates that multiple risk factors often occur together, significantly increasing the likelihood of developing diabetes.</td></tr>
</table>

Query Output:

high_risk_count
1281

3.4. Top Age Groups by Diabetes Prevalence

Analytical question: Which BRFSS age brackets have the highest diabetes prevalence?

<table>
<tr><th>SQL Query</th><th>Insight</th></tr>
<tr><td>

```
scala> spark.sql("""
     | SELECT Age,
     |         AVG(Diabetes) AS diabetes_rate
     | FROM diabetes_data
     | GROUP BY Age
     | ORDER BY diabetes_rate DESC
     | LIMIT 5
     | """).show()
+----+-------------------+
| Age|      diabetes_rate|
+----+-------------------+
|11.0| 0.4891723793526127|
|12.0|0.46837006696573696|
|10.0|  0.468082915780081|
|13.0| 0.4076410378481314|
| 9.0| 0.4059512589201562|
+----+-------------------+
```

</td><td>The highest diabetes prevalence is observed in the oldest age groups (ages 11–13), with rates exceeding 40%. This reinforces the strong correlation between aging and diabetes risk, highlighting the importance of targeted prevention and monitoring strategies for older populations.</td></tr>
</table>

Query Output:

age	diabetes_rate
11	48.9%
12	46.8%
10	46.8%
13	40.8%
9	40.6%

3.5. BMI Category Distribution

Analytical question: How are individuals distributed across standard WHO BMI categories, and what share falls in elevated-risk ranges?

<table>
<tr><th>SQL Query</th><th>Insight</th></tr>
<tr><td><pre>
scala> spark.sql("""
     | SELECT BMI_Category,
     | COUNT(*) AS count
     | FROM diabetes_data
     | GROUP BY BMI_Category
     | ORDER BY count DESC
     | """).show()
+------------+-----+
|BMI_Category|count|
+------------+-----+
3	84661
2	82859
1	58927
0	3053
+------------+-----+
</pre></td><td>The highest diabetes prevalence is observed in the oldest age brackets represented by Age codes 10 to 13, with the highest rate appearing in Age code 11 at 48.91%. These are ordinal codes from the BRFSS dataset, not literal ages — Age code 11 = 70–74 years, Age code 12 = 75–79 years, Age code 13 = 80 or older. The results further support the strong correlation between increasing age and diabetes risk.</td></tr>
</table>

Query Output:

bmi_category	total_individuals
Obese	84,661

Overweight	82,859
Normal	58937
Underweight	4,899

4. Summary

The SQL-based analysis revealed the following key patterns:

- Diabetes prevalence increases consistently across higher age brackets, rising from 6.92% in the 18–24 bracket (Age code 1) to 40.03% in the 80+ bracket (Age code 13), confirming age as a primary demographic risk factor.
- Individuals with prediabetes or diabetes have an average BMI of 31.86 compared to 27.96 for non-diabetic individuals, reinforcing the strong link between obesity and diabetes.
- A total of 1,281 individuals were identified with five simultaneous risk factors (high blood pressure, smoking, physical inactivity, no fruit consumption, no vegetable consumption), demonstrating that risk factors frequently co-occur.
- The top five age brackets by diabetes prevalence all represent individuals aged 60 and older (Age codes 9–13), further confirming the BRFSS ordinal encoding is correctly interpreted.
- Over 73% of the dataset population falls in the overweight or obese BMI categories, supporting BMI as a key feature for downstream machine learning classification.

5. Conclusion

This phase demonstrated the use of Spark SQL and DataFrames to analyse the Diabetes Health Indicators dataset efficiently. Five non-trivial SQL queries were designed, each addressing a specific analytical question. Query text is fully included in this report to allow verification of correctness and non-triviality.

The results showed that diabetes prevalence increases with age (using the correct BRFSS ordinal Age codes 1–13, representing brackets from 18–24 to 80+) and is strongly associated with higher BMI values.

The high-risk filter query identified 1,281 individuals with five coexisting risk factors, and the BMI distribution query used a window function to compute percentages efficiently in a single pass. These findings provide a solid analytical foundation for the machine learning phase.

6. References

[1] Centers for Disease Control and Prevention, "About Adult BMI," 2023. [Online]. Available: https://www.cdc.gov/bmi/adult-calculator/index.html. [Accessed: Feb. 27, 2026].